

ORLANDO EXECUTIVE, FL, USA

WMO: 722053

Lat: 28.545N

Lon: 81.333W

Elev: 108

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.1	43.3	19.4	14.7	47.5	25.2	19.3	50.0	23.3	62.9	20.5	63.8	7.4	340	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.9	93.7	76.2	92.5	76.1	91.1	75.9	80.0	86.8	79.2	86.1	78.4	85.5	7.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	148.9	82.3	77.4	143.1	81.6	76.8	140.0	81.3	43.3	86.1	42.6	85.9	41.9	85.8	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.7	17.7	16.0	DB	32.6	96.5	4.0	2.0	29.8	98.0	27.5	99.1	25.2	100.2	22.3	101.7	
			WB	28.3	81.6	3.7	1.8	25.6	82.9	23.4	84.0	21.3	85.0	18.6	86.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	73.6	60.9	64.3	68.0	72.9	78.2	81.7	83.2	81.5	76.3	68.7	64.2
		DBStd	9.54	8.21	7.70	6.63	5.16	3.87	2.86	2.17	2.08	2.32	5.04	7.98
		HDD50	26	15	5	0	0	0	0	0	0	0	0	6
		HDD65	488	176	97	49	9	1	0	0	0	5	36	115
		CDD50	8649	352	405	558	686	875	950	1028	944	815	562	446
		CDD65	3636	48	78	142	245	410	500	563	563	494	355	148
		CDH74	31334	239	461	941	1824	3547	4647	5715	5715	4496	2574	760
		CDH80	11276	15	74	196	538	1407	1913	2357	2265	1617	731	119
(14)	Wind	WSAvg	7.2	7.4	7.7	8.3	8.2	7.6	6.7	6.0	6.1	6.6	7.2	7.1
(15)	Precipitation	PrecAvg	51.0	2.4	2.4	3.5	2.3	3.4	7.8	7.3	7.5	6.7	3.7	1.8
		PrecMax	68.3	5.8	8.8	11.3	7.2	17.0	17.3	12.9	18.0	14.0	10.2	7.9
		PrecMin	33.3	0.1	0.4	0.0	0.1	0.4	1.7	2.7	3.2	3.3	0.5	0.4
		PrecStd	9.1	1.6	1.8	2.9	1.4	3.1	3.1	2.6	3.3	2.8	2.3	1.7
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.9	85.1	87.4	90.4	93.7	97.0	95.4	94.9	92.9	90.3	86.0
			MCWB	69.2	69.8	70.8	71.0	72.6	76.3	77.3	77.2	76.0	76.8	73.2
		2%	DB	79.4	82.2	84.4	87.9	91.3	93.4	93.2	93.1	91.0	88.3	83.3
			MCWB	67.5	68.5	68.8	70.4	72.1	76.0	76.9	77.0	75.9	75.3	71.9
		5%	DB	76.9	79.9	82.1	85.6	89.9	91.4	91.8	91.5	89.9	86.3	81.1
			MCWB	66.2	67.1	67.5	69.2	72.0	75.7	76.8	77.1	75.9	74.0	70.0
		10%	DB	73.5	76.7	79.7	83.2	87.8	89.7	90.4	90.2	88.2	84.1	78.8
			MCWB	65.1	65.7	66.1	68.3	71.5	75.4	76.9	77.2	75.8	73.0	68.3
		0.4%	WB	72.1	72.4	74.0	75.2	77.3	79.8	81.0	81.0	80.5	79.6	75.9
			MCDB	78.0	79.1	81.2	83.5	86.0	89.0	88.3	88.0	85.4	85.1	81.7
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	70.9	72.0	73.5	76.0	78.4	79.6	79.9	79.3	78.0	74.2
			MCDB	75.6	78.0	79.4	81.8	84.4	87.1	87.0	86.8	84.9	83.5	79.5
		2%	WB	68.4	69.5	70.4	72.3	75.0	77.8	78.7	79.1	78.4	76.8	72.7
			MCDB	74.1	76.3	77.7	80.4	83.3	86.2	86.3	86.0	84.3	82.3	77.7
		5%	WB	66.7	68.0	68.9	71.0	74.0	77.1	78.1	78.4	77.7	75.8	71.0
			MCDB	72.2	74.6	76.1	79.0	82.0	85.1	85.7	85.3	83.9	81.2	76.2
		10%	WB	66.7	68.0	68.9	71.0	74.0	77.1	78.1	78.4	77.7	75.8	71.0
			MCDB	72.2	74.6	76.1	79.0	82.0	85.1	85.7	85.3	83.9	81.2	76.2
		0.4%	WB	66.7	68.0	68.9	71.0	74.0	77.1	78.1	78.4	77.7	75.8	71.0
			MCDB	72.2	74.6	76.1	79.0	82.0	85.1	85.7	85.3	83.9	81.2	76.2
(35)	Mean Daily Temperature Range	MDBR	18.8	19.0	19.3	19.3	18.1	15.9	15.5	14.9	14.1	15.0	16.9	17.4
		5% DB	MCDBR	19.6	20.4	20.4	20.4	19.9	18.2	16.8	16.3	15.7	15.5	17.1
			MCWBR	9.7	8.6	8.0	7.0	5.8	5.4	5.2	5.0	4.7	5.4	8.1
		5% WB	MCDBR	17.5	17.7	17.6	18.4	17.3	16.1	15.6	14.9	13.9	13.4	14.6
			MCWBR	10.5	9.1	9.0	8.0	6.5	5.7	5.6	5.5	5.2	5.5	7.7
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.352	0.373	0.391	0.425	0.455	0.476	0.465	0.421	0.396	0.357	0.346
		taud	2.525	2.496	2.428	2.356	2.309	2.298	2.236	2.284	2.398	2.433	2.496	2.526
		Ebn at Noon	287	292	291	287	276	266	260	263	272	272	277	278
		Edh at Noon	29	32	36	40	42	42	45	42	36	33	29	27
(44)	All-Sky Solar Radiation	RadAvg	1082	1282	1625	1927	2000	1797	1798	1708	1534	1397	1148	987
		RadStd	80	84	123	99	134	167	105	114	88	98	62	77

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-86	+320	N/A
(47)	Regional (44 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	-9	-85	+288	+209

Nomenclature: See separate page